higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE

## QUANTITY SURVEYING N4

## 14 AUGUST 2019

**This marking guideline consists of 8 pages.**

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## SECTION A

## QUESTION 1

**1.1.1**

2 / 2 / 0,5 — 5,50 × 0,20 ✓
5,50 × 1,66 ✓

220 mm thk. bk. wl in gable ends a.b. ✓

Rectangle

Triangle (3)

**1.1.2**

2 ✓ — 11,06 × 0,19

110 mm thk. bk. wl. in bm. filling. ✓

11,500 – 2/0,220
= 11,060 m

(2)

**1.1.3**

2 / 2 / 0,5 — 5,50 × 0,20 ✓
5,50 × 2 / 1,66 ✓
11,06 × 0,19 ✓

E.o. ord. bks. a.b. for fcgs. ✓

- Rectangle
- Triangle
- Bm. fill (4)

**1.2.1**

2 — 11,06 ✓

38 x 114 mm tbr. w.p. ✓

(2)

**1.2.2**

2 — 11,06 × 0,30 ✓

Treat tbr. w.p. w. carb. oil. ✓

2/114 + 2/38 = 0,304 m

(2)

(1)

**1.3.1**

11,060 ÷ 1,200 = 9,217

19 + 1 = 10 No. ✓

(1)

**1.3.2**

All in 10 No. nailed full trusses

10 / 2 / 10 / 2 — 1,86 ✓ ; 1,66 ✓

38 x 114 mm S.S.A.P. tbr. n.e. 2,4 m long. ✓

- Strut
- Post

10 / 2 / 10 / 2 — 4,20 ✓ ; 5,50 ✓

Ditto, but exce. 3,9 m n.e. 6,6 m long. ✓

-Rafters

-Tie

(6)

**1.3.3**

2 / 4 — 11,90 ✓

50 x 76 mm S.S.A.P. tbr purlins. ✓

11,500 + 2/0,200 = 11,900 m

(2)

**1.3.4**

2 — 11,90

38 x 200 tbr. fascia brd. ✓

(1)

(2)

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | Roof covering<br>1.4.1 | | | | |
| 2 | | | | | | | |
| | 11,90 | ✓ | | | | | |
| | 4,20 | ✓ | F. c. roof shtg. fixed on tbr. purlins. ✓ | | | | |
| | | | (3) | | | | |
| | | | 1.4.2 | | | | |
| | 11,9 0 | ✓ | F.c. ridge capping. ✓ | | | | |
| | | | Extra for f.c. ridge s. e. ✓ | | | | |
| | 2 | ✓ | | | | | |
| | | | (4) | | | | |

**[30]**

**QUESTION 2.1**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | Roof covering<br>1.4.1 | | | | |
| 2 | | | | | | | |
| ✓ | 11,90 | ✓ | | | | | |
| | 4,20 | ✓ | F. c. roof shtg. fixed on tbr. purlins. | | | | |
| | | 99,96 | | | | | |
| | | | ✓ ✓ | | | | |
| | | | 1.4.2 | | | | |
| | | | F.c. ridge capping. | | | | |
| | 11,90 | | | | | | |
| | | 11,90 | | | | | |
| | | | (3) | | | | |
| | | | | | 2 | 2 | Extra for f.c. ridge s. e. |
| | | | | | | | (4) |

(½ × 5) (2½)

## QUESTION 2.2

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | ✓ | ROOF CONSTRUCTION | | | | |
| ROOF COVERING | | ✓ | | | | | | |
| ✓ | | | | | | | | |
| M² | | ✓ | | | | | | |
| F.c. rf. shtg. fixed on tbr. purlins | | | | | | | | |
| 99,96 | 3 | ✓ | | | | | | |
| ✓ | | | | | | | | |
| 100 M² | | ✓ | | | | | | |
| ✓ | | ✓ | | | | | | |
| M | | | | | | | | |
| F.c. ridge capping | | | | | | | | |
| 11,90 | 3 | | | | | | | |
| 12 M | | | | | | | | |
| No | | | | | | | | |
| F.c. ridge s.e. | | | | | | | | |
| 2 | 4 | | | | | | | |
| 2 No | | | | | | | | |

(½ × 7) (3½)

**QUESTION 2.3**

| | | | | | |
|---|---|---|---|---|---|
| | ROOF CONSTRUCTION ✓ | | | | |
| | Bill No. ✓ | | | | |
| | Roof covering ✓ | | | | |
| ✓ | ✓ | ✓ | ✓ | | |
| 1. | Fibre cement roof sheeting on timber purlins | $M^2$ | 100 | | |
| 2. | Fibre cement ridge capping | M | 12 | | |
| 3. | Extra for fibre cement ridge stopped ends | No | 2 | | |
| | Carried to summary ✓ | | | R | |

(0,5 × 8) (4)

Working up mark allocation

| | | |
|---|---|---|
| Squaring | 0,5 | |
| Check dimensioning. (ticks) | 0,5 | |
| Square dimensions. | 0,5 | |
| Check squaring answers. (ticks) | 0,5 | |
| Transfers to abstract. | 0,5 | (2,5) |
| Abstracting | | |
| Section of work | 0,5 | |
| Trades | 0,5 | |
| Units | 0,5 | |
| Page referencing | 0,5 | |
| Use of columns | 0,5 | |
| Reducing | 0,5 | |
| Transfers to bill | 0,5 | (3,5) |
| Billing | | |
| Section of work | 0,5 | |
| Bill no. | 0,5 | |
| Trade | 0,5 | |
| Item no. | 0,5 | |
| Descriptions (no abbreviations) | 0,5 | |
| Units | 0,5 | |
| Quantities | 0,5 | |
| Summary | 0,5 | (4) |

**[10]**

**TOTAL SECTION A: 40**

## SECTION B

## QUESTION 3

3.1 A list drawn up that indicate the sections✓ of the work that must be measured✓

3.2 A list drawn up by the measurer of all the items✓ in a section of work to be measured✓

3.3 A list drawn up by a measurer of uncertain issues✓ to be cleared with the architect✓

3.4 A list drawn up of all the contractors✓ who are eligible or invited to submit a tender price✓

3.5 A list of all the drawings that were used✓ to prepare the bills of quantities✓

(5 × 2) **[10]**

## QUESTION 4

4.1
- Check all dimensions taken from the drawings (given and scaled).
- Check all the waste calculations for correctness.
- Check all the side casts for correctness.
- Check all the dimensions inserted in the dimension columns.
- Check the squaring of the dimensions.
- Check the answers.
- Check the final answers. (Any 5 × 1) (5)

4.2
- Check the transfers from the dimension sheets to the abstract sheets.
- Check the section of work measured.
- Check the trade.
- Check the unit.
- Check the page reference.
- Check the casting up.
- Check the reduction. (Any 5 × 1) (5)

**[10]**

## QUESTION 5

5.1
5.1.1 As usual
5.1.2 Exceeding
5.1.3 Rainwater pipe
5.1.4 Damp-proof course
5.1.5 Striking off and cure

(5 × 1) (5)

5.2
- It must be measured in cubic metres.
- It must be measured as an extra over all excavations.
- It must be dumped at a prescribed place.
- The location and
- the approximate distance must be stated. (5)

**[10]**

## QUESTION 6

6.1 Item no., description, quantity, rate, amount (5)

6.2
- Horizontal at right angles to the line of sight
- Horizontal parallel to the line of sight
- Vertical (3)

6.3
- Fractional quantities must be rounded off to the nearest whole quantity.
- Quantities of a significant value may be taken to one decimal. (2)

**[10]**

## QUESTION 7

7.1 7.1.1 Duties performed before a contractor is appointed

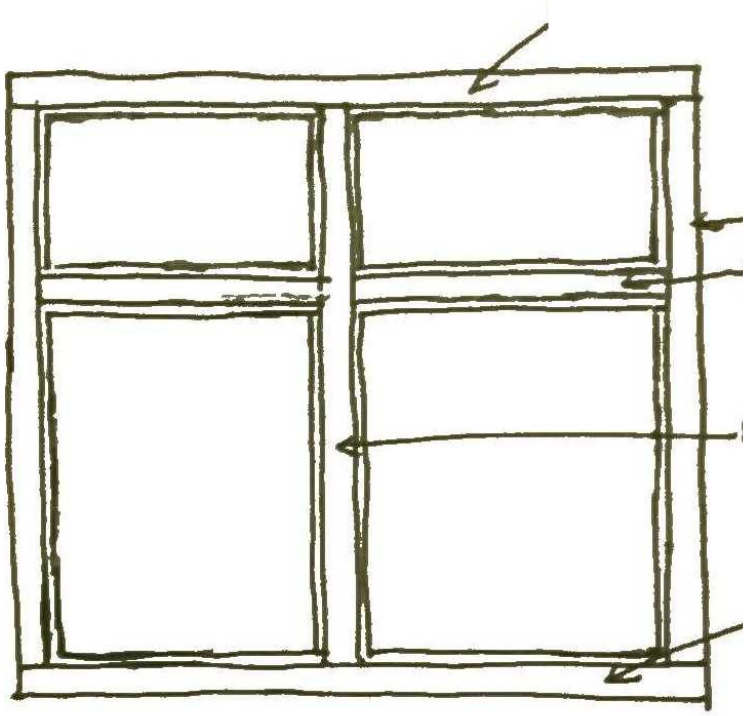
7.1.2 Duties performed after construction work commenced (2 × 2) (4)

7.2
- Prepares costing estimates and budgets
- Prepares bills of quantities
- Writes specifications
- Gives advice of new building methods and materials
- Gives advice about tender methods
- Calls for tenders (Any 3 × 1) (3)

7.3
- Checks priced bills of quantities
- Interim valuations
- Remeasures provisional quantities
- Cost variations
- Final accounts (Any 3 × 1) (3)

**[10]**

**QUESTION 8**



Head

Stile

Transom

Mullion

Sill

**TIMBE** (5 × 2) **V** **[10]**

**TOTAL SECTION B: 60**
**GRAND TOTAL: 100**

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